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Che-50

Chapter - 1

2022(t); TU, 2017, 2015

(*) Define drift speed, ionic mobility & transport number?

⇒ Drift speed: The average velocity attained by charged particles, such as e^- in a material due to an electric field is called drift speed.

$$V_d = \frac{I}{neA}$$

where, I = current flow
n = free e^- density
A = cross sectional area.

⇒ ionic mobility: It refers to the speed at which an ion moves through a solution or other medium when subjected to an electric field. It is a measure of how quickly an ion can travel under the influence of the field and is usually expressed in units of m^2/Vs. Ionic mobility depends on factors such as the size & charge of the ion, the viscosity of the medium & the temperature.

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⇒ transport number: It refers to the fraction of the total electrical current carried by a particular ion in an electrolyte solution. The t. numbers quantifies the contribution of each ion to the total current.

transport number of cation, $t_+ = \frac{V_+}{V_- + V_+}$

transport number of anion, $t_- = \frac{V_-}{V_- + V_+}$

The sum of the transport numbers of all ions in the sln is equal to 1.

$$t_+ + t_- = 1$$

2017, 2015

Define (i) mean free path (ii) collision frequency?

⇒ (i) Mean free path is the average distance a particle, such as a molecule, atom or e⁻ travels btwn successive collisions with other particles in a medium.

$$\lambda = \frac{KT}{\sqrt{2}\,\pi d^2 n}$$

where, n = number of molecules per unit volume.

d = diameter of the molecules.

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(ii) collision frequency refers to the number of collisions that occur per unit time btwn particles, such as molecules or atoms in a medium.

collision frequency, $z = \sigma \bar{C}_{rel} N$ ---------- (1)

$$N = \frac{N}{V} = \frac{nN_A}{V} = \frac{PN_A}{RT} = \frac{P}{KT}$$

$$\therefore z = \frac{\sigma \bar{C}_{rel} P}{KT}$$

where, $\sigma = \pi d^2$ which is area, N = number density

$\bar{C}_{rel}$ = relative mean speed.

2018, 2021, 2015

(8) Explain transport properties of perfect gases?

⇒ Transport properties include viscosity, thermal conductivity and diffusion.

1. Viscosity: It describes how the molecules of the gas collide with each other & exchange momentum. In an ideal gas, viscosity depends on the molecular size, mass & the speed of the molecules. As T increases, the viscosity of a perfect gas also increases.

2. Thermal conductivity: It measures the ability of a

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gas to conduct heat. In perfect gases, it depends on T. & molecular motion. As T. rises, molecules move faster, leading to more effective energy transfer and higher thermal conductivity.

3. Diffusion: It describes the rate at which molecules of a gas spread out from regions of higher to lower concentration. In perfect gases, Diffusion occurs through random molecular motion. The diffusivity depends on the molecular size, mass & Temp. As T. increases, Diffusion increases because the gas molecules more more rapidly.

2022, 2021, 2018, 2017, 2022(t), 2015

⊗ State & explain Graham's law of ~~effusion~~?

⇒ It states that the rate of effusion of a gas is inversely proportional to the square root of its molar mass.

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

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The law can be used to determine the molar mass of an unknown gas by comparing its effusion rate. effusion is the process by which gas molecule escape through a small hole in a container without collision with other molecules. According to the law,

$$r \propto \frac{1}{\sqrt{M}}$$

$$\Rightarrow r = K \frac{1}{\sqrt{M}} \quad \text{.....(i)}$$

where "K" is the proportionality constant. For two different gases eqn (i) becomes:

$$r_1 = K \frac{1}{\sqrt{M_1}} \quad \text{----(ii)}$$

$$r_2 = K \frac{1}{\sqrt{M_2}} \quad \text{----(iii)}$$

Dividing eqn (i) by (ii) we get,

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

which is Graham's law of effusion

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2022, 2018, 2016

⊛ What is meant by free path? Derive an eqn for the mean free path?

⇒ Free path refers to the actual distance that a specific particle travels btwn two successive collision.

When a gas molecule collides with the walls of a container with a frequency z, it spends a time $\frac{1}{z}$ btwn collision & therefore travel a distance $\left(\frac{1}{z}\right)\bar{c}$.

thus the mean free path is,

$$\lambda = \left(\frac{1}{z}\right)\bar{c} \quad \text{----} \quad (1)$$

where, $\bar{c}$ is the mean speed.

From collision frequency we know that,

$$z = \frac{\sigma \bar{C}_{rel} P}{KT} = \frac{\sigma \sqrt{2}\, \bar{c}\, P}{KT} \quad \text{----} \quad (11) \quad \left[\because \bar{C}_{rel} = \sqrt{2}\,\bar{c}\right]$$

where, $\bar{C}_{rel}$ = relative mean speed, σ = collision cross section of molecule

From (1) & (11),

$$\lambda = \frac{KT}{\sigma \sqrt{2}\, \bar{c}\, P} \times \bar{c}$$

$$\therefore \lambda = \frac{KT}{\sqrt{2}\, \sigma P}$$

which is the expression for mean free path.

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2016
Calculate the mean free path of molecules in air using $\sigma = 0.93\ nm^2$ at 25°C and the pressure is

(i) 10 atm (ii) 1 atm.

⇒ (i) we know,

$$\lambda = \frac{KT}{\sqrt{2}\,\sigma P}$$

$$= \frac{1.38 \times 10^{-23} \times 298}{\sqrt{2} \times 0.93 \times 10^{-18} \times 10 \times 101325}$$

$$= 6.67 \times 10^{-9}\ m$$ Ans

(ii) $\lambda = \frac{1.38 \times 10^{-23} \times 298}{\sqrt{2} \times 0.93 \times 10^{-18} \times 101325} = 6.67 \times 10^{-8}\ m$ Ans.

2016
The mobility of a chloride ion in aq. s/n at 25°C is $7.91 \times 10^{-8}\ m^2 s^{-1} V^{-1}$. Calculate its molar ionic conductivity?

⇒ molar ionic conductivity $\propto$ mobility

$$\lambda = z \mu F$$

Where, z = ionic charge, μ = mobility, F = faraday's constant = 96500 $C mol^{-1}$

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$$\lambda = z\mu F = 1 \times 7.91 \times 10^{-8} \times 96500$$

$$= 7.63 \times 10^{-3} \text{ mS m}^2 \text{ mol}^{-1}.$$

(2016) The mobility of negative ion 1:1 electrolyte in aq. soln at 25°C is $6.85 \times 10^{-8} \text{ m}^2 \text{s}^{-1} \text{V}^{-1}$. calculate its molar ionic conductivity?

$\Rightarrow$

$$\lambda = z\mu F = 1 \times 6.85 \times 10^{-8} \times 96500$$

$$= 6.61 \times 10^{-3} \text{ mS m}^2 \text{ mol}^{-1}.$$

(2017) The limiting molar conductivities of KCl, KNO_3 and $AgNO_3$ are 14.99, 14.50 & 13.34 $\text{ms m}^2 \text{mol}^{-1}$. what is the limiting molar conductivity of AgCl?

$\Rightarrow$ A. to Kohlrausch law,

$$\Lambda_{KCl} = \lambda^\circ_{K^+} + \lambda^\circ_{Cl^-} = 14.99 \quad \text{---} (1)$$

$$\Lambda_{KNO_3} = \lambda^\circ_{K^+} + \lambda^\circ_{NO_3^-} = 14.50 \quad \text{---} (II)$$

$$\Lambda_{AgNO_3} = \lambda^\circ_{Ag^+} + \lambda^\circ_{NO_3^-} = 13.34 \quad \text{---} (III)$$

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Now, ① + ③ − ②

$$\lambda^\circ_{K^+} + \lambda^\circ_{Cl^-} + \lambda^\circ_{Ag^+} + \lambda^\circ_{NO_3^-} - \lambda^\circ_{K^+} - \lambda^\circ_{NO_3^-} = 14.99 + 13.34 - 14.50$$

$$\Rightarrow \lambda^\circ_{Ag^+} + \lambda^\circ_{Cl^-} = 13.83 \text{ mS m}^2\text{mol}^{-1}$$

$$\therefore \Lambda_{AgCl} = 13.83 \text{ mS m}^2\text{mol}^{-1} \quad \text{Ans}$$

(✗) 2022 $T_{1/2} = 20$ min. calculate required time for complete 75% of the reaction?

$$\Rightarrow \lambda = \frac{0.693}{T_{1/2}} = 0.03465 \text{ min}^{-1}$$

Now, $\frac{N}{N_0} = e^{-\lambda t} \Rightarrow \ln\left(\frac{25}{100}\right) = -\lambda t$

$$\Rightarrow t = 40\text{min} \quad \text{Ans.}$$

(✗) 2021 calculate the molar mass of a gas whose diffusion rate is 2.92 times of the diffusion rate of NH_3?

⇒ let, molar mass of required gas is M_1

Given, gas is NH_3 whose $M_{NH_3} = 17$ g.

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A. to. question,

$$r_1 = 2.92\, r_{NH_3}$$

$$\Rightarrow \frac{r_1}{r_{NH_3}} = 2.92$$

we know, $\frac{r_1}{r_{NH_3}} = \sqrt{\frac{M_{NH_3}}{M_1}}$

$$\therefore \sqrt{\frac{M_{NH_3}}{M_1}} = \frac{r_1}{r_{NH_3}} = 2.92$$

$$\Rightarrow \frac{M_{NH_3}}{M_1} = (2.92)^2 \quad \Rightarrow M_1 = 5.8\ gm$$

$\therefore M_1 = 1.99$

$\therefore M_1 \approx 2\ gm$

$\therefore$ required gas is H_2 gas.

(★) 2015 The limiting conductivities of NaI, $NaCH_3CO$ and $Mg(CH_3CO_2)_2$ are 12.69, 9.10 & 18.78 mSm^2mol^{-1} at 25°C. What is the molar conductivities of MgI_2?

$$\Rightarrow \Lambda_{NaI} = \lambda^\circ_{Na^+} + \lambda^\circ_{I^-} = 12.69 \quad ----- (i)$$

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$\Lambda_{CH_3COONa} = \lambda^\circ_{Na^+} + \lambda^\circ_{CH_3COO^-} = 9.10$ ---- (II)

$\Lambda_{Mg(CH_3CO_2)_2} = \lambda^\circ_{Mg^{2+}} + 2\lambda^\circ_{CH_3COO^-} = 18.78$ ---- (III)

Now, 2X(I) + (III) − 2X(II)

$2\lambda_{Na^+} + 2\lambda_{I^-} + \lambda_{Mg^{2+}} + 2\lambda_{CH_3COO^-} - 2\lambda_{Na^+} - 2\lambda_{CH_3COO^-} = 2X12.69 + 18.78 - 2X9.10$

$\Rightarrow \lambda_{Mg^{2+}} + 2\lambda_{I^-} = 25.96$

$\therefore \Lambda_{MgI_2} = \lambda_{Mg^{2+}} + 2\lambda_{I^-} = 25.96 \ mS\ m^2\ mol^{-1}$ Ans.

2020 Calculate value of Ea at 273 & 298 K with $K_1 = 4.1 \times 10^3$ and $K_2 = 79 \times 10^3\ min^{-1}$?

$\Rightarrow \log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$

$\Rightarrow E_a = 2.303 \times 8.314 \times \left(\frac{273 \times 298}{298 - 273}\right) \times \log\left(\frac{79 \times 10^3}{4.1 \times 10^3}\right)$

$\therefore E_a = 80025.84\ J$ Ans.

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2017

If equal amount of He & Ar are placed in a porous container and allowed to escape, which gas will escape faster & how much faster?

⇒ He escape faster than Ar.

$$\frac{r_{He}}{r_{Ar}} = \sqrt{\frac{M_{Ar}}{M_{He}}} = \sqrt{\frac{40}{4}} = 3.16$$

$\therefore r_{He} = 3.16\, r_{Ar}$

∴ He escape 3.16 times faster than Ar.

2021

The rate constant of fast order reaction at 25°C is $0.24\ s^{-1}$. If $E_a = 88\ KJmol^{-1}$. at what T. $K = 4 \times 10^{-2}\ s^{-1}$?

$$\Rightarrow \log\left(\frac{K_2}{K_1}\right) = \frac{E_a}{2.303R} \times \left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\Rightarrow \frac{T_2 - T_1}{T_1 T_2} = \frac{2.303 \times 8.314}{88000} \times \log\left(\frac{4 \times 10^{-2}}{0.24}\right)$$

$$\Rightarrow T_2 - T_1 = -0.000169\, T_1 T_2$$

$$\Rightarrow T_2 + 0.05 T_2 = 298 \quad \therefore T_2 = 283.81\ K$$

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2022, 2020, 2017

(★) Derive an eqn for collision theory of gas?

⇒ collision theory postulates that for a reaction to take place, reactant molecules must collide with sufficient energy & correct orientation.

For gas molecule A & B with mases m_A & m_B, their pespective speed (r.m.s) are:

$$U_A = \sqrt{\frac{3KT}{m_A}} \quad , \quad U_B = \sqrt{\frac{3KT}{m_B}}$$

The relative speed of two molecules is

$$U_{AB} = \sqrt{U_A^2 + U_B^2} = \sqrt{\frac{3KT}{m_A} + \frac{3KT}{m_B}} = \sqrt{3KT\left(\frac{1}{m_A} + \frac{1}{m_B}\right)}$$

collision frequency, $Z_{AB} = N_A N_B \bar{\sigma}_{AB} U_{AB}$

where, N_A & N_B are the number densities (molecules per unit volume) of gas A & B respectively.

σ_{AB} is the collision cross-section, U_{AB} is the relative speed btwn molecules A & B.

Not all collisions result in a reaction. For a reaction to occur, the colliding molecules must have energy

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greater than E_a. The fraction of molecules with energy greater than E_a is given by the Boltzman distribution:

$$f = e^{-\frac{E_a}{RT}}$$

The rate R is proportional to the product of collision frequency & the fraction of collisions that overcome the activation energy:

$$R \propto Z_{AB} \cdot e^{-\frac{E_a}{RT}}$$

$$\Rightarrow R = K Z_{AB} e^{-\frac{E_a}{RT}}$$

$$\therefore R = K N_A N_B \sigma_{AB} U_{AB} e^{-\frac{E_a}{RT}}$$

This eqn represent the collision theory for gas-phase reactions.

(2017) Derive an eqn for collision frequency?

⇒ collision frequency z, is the number of collisions made by one molecule divided by the time.

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interval during which the collisions are counted, when there are N molecules in a volume V is

$$Z = \sigma \bar{C}_{rel} N$$

where, N = Number density $= \frac{N}{V} = \frac{n N_A}{V} = \frac{P N_A}{RT} = \frac{P}{KT}$

$$\therefore Z = \frac{\sigma \bar{C}_{rel} P}{KT}$$

2016, 2015, 2022

Write short note on Kohlrausch law?

⇒ Kohlrausch state that the molar conductance at infinite dilution is different for different electrolytes & is equal to the sum of the conductance of the constituents ions of the electrolytes

$$\Lambda_m = \lambda^\circ_+ + \lambda^\circ_-$$

where λ°_+ & λ°_- are called the ion conductance of the cation & anion respectively. this eqn means that, each ion of an electrolyte contributes independently to the value of molar conductance

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at infinite dilution. Kohlrausch's law helps in calculating the dissociation constant of weak electrolytes by measuring their molar conductivity at different concentration. By this law one can calculate limiting molar conductivity. It provides insights into the mobility of ions in sln. For example "the limiting molar conductivity of electrolyte NaCl is the sum of limiting molar conductivities of Na^+ ion and Cl^- ion.

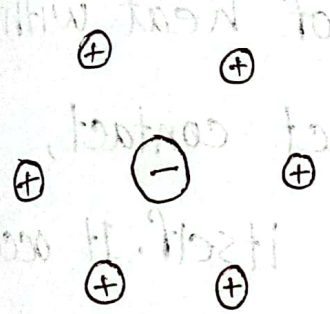
$$\Lambda_{NaCl} = \lambda^\circ_{Na^+} + \lambda^\circ_{Cl^-}$$

2022, 2016
Explain relaxation effect and electrophoric effect in reduction of ions mobility?

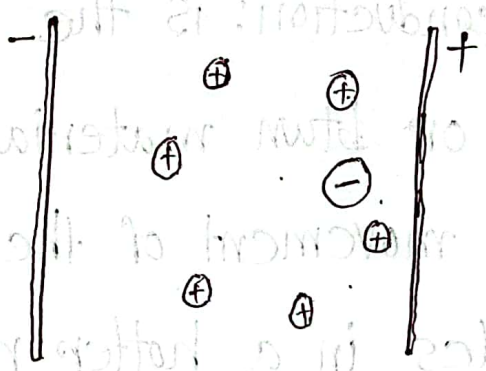
⇒ relaxation effect: when the external field is applied, the central ion starts moving toward the electrode of opposite charge and more of

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the ionic atmosphere is left behind than is present on the front side. These excess ions of the ionic atmosphere which are left behind will tend to retard the speed of the moving ion due to electrostatic force. This effect is called relaxation effect.



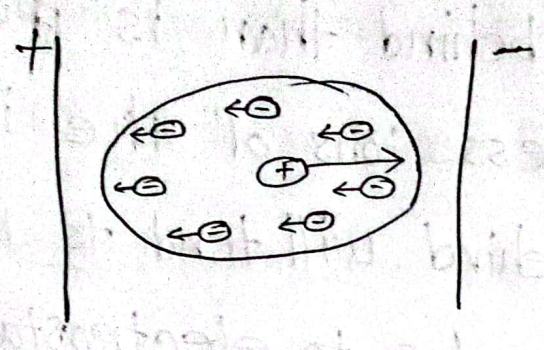
symmetrical ionic atmosphere at rest (no electric field)



assymetrical ionic atmosphere under the influence of applied field.

electrophoretic effect: The e. effect is caused by the tendency of the applied electric field to move the ionic atmosphere along with the associated solvent molecules in a direction opposite direction in which the central ion is moving. It may well be imagined that the central ion is moving against the flow of the ionic atmosphere with reduced speed. This effect is called electrophoretic effect.

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2015+TU

⊛ Define thermal conduction & collusion flux?

⇒ Thermal conduction: is the transfer of heat within a material or btwn materials in direct contact, without any movement of the substance itself. It occurs when particles in a hotter region transfer Kinetic energy to neighbouring particles in a cooler region, thus equalising the temp. over time.

collision flux is a measure of the number of collisions per unit area, per unit time, & per unit concentration. collisio flux z can be expressed as:

$$z = n V_{avg} \cdot \sigma$$

where, n = number density (number of particle per unit volume)

V_{avg} = average relative velocity btwn particles.

σ = collision cross-section.

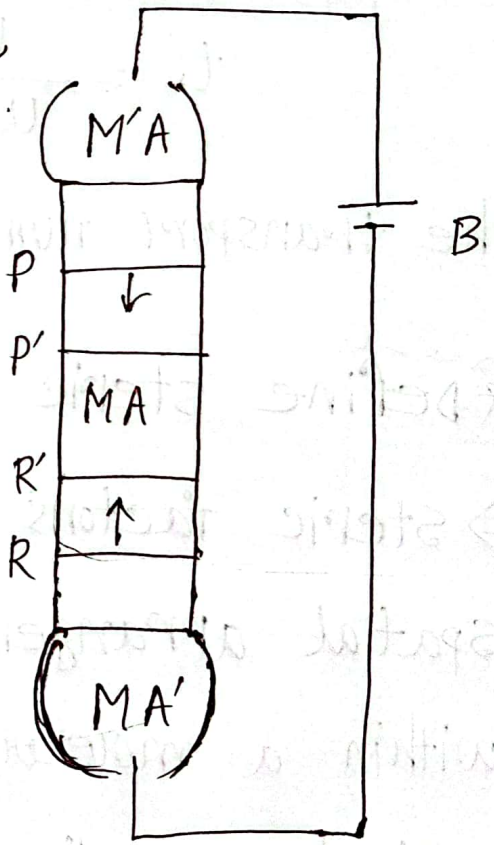
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2022, 2015, 2017

⊛ Describe a suitable method for measurment of transport numbers ?

⇒ Moving boundary method: This method is based on the principle that, transport number can be calculated if cationic & anionic

Let the sln of an electrolyte MA, which is to be studied, is placed btwn the sln of two other salts MA' & M'A such that M'A has the anion A common with MA while MA' has the cation M common with MA. This is essential to maintain sharp boundaries btwn the three sln of electrolytes. the initial sharp boundary btwn the sln M'A and MA is shown by the horizontal line p whereas btwn MA' & MA is shown by R. By passing current



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from the source B, ionic migration starts and boundary P moves downwards to p' while the boundary R moves to R'. The distance travelled by the two boundary in time t are pp' and RR' which are directly proportional to the cationic & anionic velocity respectively. Therefore,

$$t_+ = \frac{U_+}{U_+ + U_-} = \frac{PP'}{PP' + RR'}$$

$$t_- = \frac{U_-}{U_+ + U_-} = \frac{RR'}{PP' + RR'}$$

The transport numbers are thus measured.

2020

✱ Define steric factor?

⇒ Steric factors refer to the influence of the spatial arrangement of atoms or group of atoms within a molecule on its chemical behavior & physical properties. These factors arise from the physical size & shape of atoms which can hinder or prevent reactions due to crowding. Steric effects can impact reaction rates, stability of intermediates.

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steric factors play a crucial role in determining molecular geometry, reactivity & selectivity in chemical reactions. Arhenius eqn,

$$K = A e^{-Ea/RT}$$

$$= Z \times P \, e^{-Ea/RT}$$

Z ↑ frequency factor; P ↑ steric factor

2020

⊛ Determine the ratio of (i) mean speeds (ii) mean kinetic energies of He & Hg atoms at 25°C?

(i) we know, $V_{mean} = V_{avg} = \bar{V} = \sqrt{\frac{8RT}{\pi M}}$

∴ ratio, $\frac{V_{He}}{V_{Hg}} = \sqrt{\frac{8RT}{\pi M_{He}}} \times \sqrt{\frac{\pi M_{Hg}}{8RT}} = \sqrt{\frac{M_{Hg}}{M_{He}}} = \sqrt{\frac{200.592}{4}}$

$\therefore V_{He} : V_{Hg} = 7.08 : 1$

(ii) we know, $E_{K_{avg}} = E_{K_{mean}} = \bar{E}_K = \frac{3}{2} KT$

$\therefore \frac{E_{K_{He}}}{E_{K_{Hg}}} = \frac{3}{2} KT \times \frac{2}{3KT} = 1$

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$\therefore E_{K_{He}} : E_{K_{Hg}} = 1:1$ Ans

⊛ 2021

The mobility of a Li^+ ion in an aq. sln is $4.01 \times 10^{-8}\ m^2 s^{-1} V^{-1}$ at 25°C. The potential dfrnce btwn two electrods is 12.0 V. If electrodes are 1.00 cm apart, what is drift speed of the ion?

⇒ ionic mobility, $\mu = \frac{V_d}{E}$

$$\Rightarrow V_d = \mu E = \mu \times \frac{V}{d} = 4.01 \times 10^{-8} \times \frac{12}{0.01}$$

$$\therefore V_d = 4.81 \times 10^{-9}\ ms^{-1}$$ Ans

⊛ 2021, TU

Explain Fick's first & second law of diffusion?

⇒ First law: It states that "Movement of particles from high to low concentration is directly proportional to the particle's concentration gradient.

$$J \propto \frac{d\varphi}{dx} \quad \text{or} \quad J = -D\frac{d\varphi}{dx}$$

where, J = diffusion flux, D = diffusion coefficient, $d\varphi$ = change in concentration of the particle, dx = change in position

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$\frac{d\varphi}{dt}$ = concentration gradient of the particle.

2nd law: It state that the change in conc. w.r.to time at a particular region is proportional to the changes in conc. gradient at that point in the system.

$$\frac{d\varphi}{dt} = D\frac{d^2\varphi}{dx^2}$$

where, $d\varphi$ = changes in conc. of the particle.

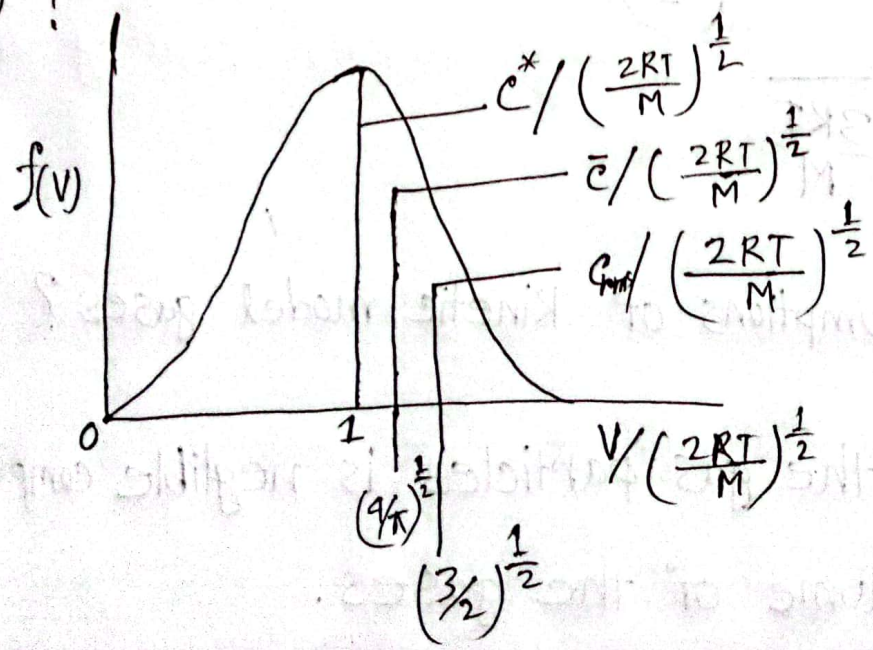
dx = changes in position, $\frac{d\varphi}{dt}$ = changes in conc. with time

$\frac{d^2\varphi}{dx^2}$ = second derivative of $\frac{d\varphi}{dx}$

2020

(*) Derive eqn for most probable velocity (V_{mp}) & average velocity (V_{avg}) ?

∴



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(i) calculation most probable speed from figure,

$$c^* / \left(\frac{2RT}{M}\right)^{1/2} = 1$$

$$\Rightarrow c_{mp} = c^* = \sqrt{\frac{2RT}{M}}$$

(ii) calculation c_{avg} from graph,

$$\bar{c} / \left(\frac{2RT}{M}\right)^{\frac{1}{2}} = \left(\frac{4}{\pi}\right)^{\frac{1}{2}}$$

$$\Rightarrow \bar{c} = c_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

(iii) calculation c_{rms},

$$c_{rms} / \left(\frac{2RT}{M}\right)^{\frac{1}{2}} = \left(\frac{3}{2}\right)^{\frac{1}{2}}$$

$$\Rightarrow c_{rms} = \sqrt{\frac{3RT}{M}}$$

(*) 2020 write the assumptions of kinetic model gases?

⇒ The volume of the gas particles is neglible compared to the total volume of the gases.

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② Gases are composed of tiny particles that are in constant random motion.

③ there are no attractive or repulsive forces btwn the gas particles.

④ collisions btwn gas particles are perfectly elastic.

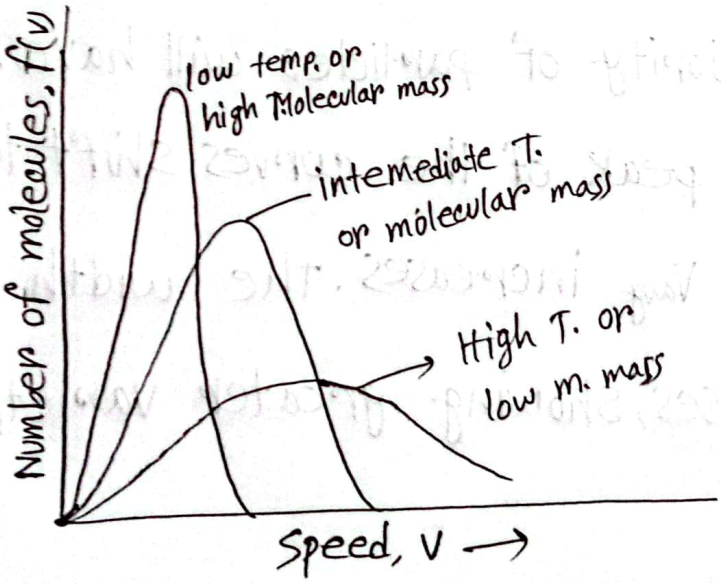
⑤ E_K is directly proportional to the absolute Temp. of the gas.

⑥ Each molecule has different velocity.

2016, 2022, 2021, 2020, TU

⑦ Draw & explain the maxwell distribution of speed?

⇒



the prabability distribution function, $f(v) = 4\pi\left(\frac{m}{2\pi KT}\right)^{3/2} v^2 e^{(-mv^2/2KT)}$

where, m is the mass of gas particle, K = Bolzman constant.

The maxwell distribution is represented by a curve

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that show the number of molecules having a certain speed plotted against the speed itself.
The curve is asymmetrical, with a long tail extending to higher speeds. It peaks at a specific speed, representing the most probable speed. this is the speed at which the largest number of molecules are travelling. At higher Temp. V_{avg} & V_{mp} both increases.

$$V_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2KT}{m}}$$

This eqn indicates at a given temp. there is a specific speed that the majority of particles will have. As Temp. increases the peak of the curves shift to the right, indicating that V_{avg} increases. the width of the distribution also increases, showing greater variety of speeds.

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2022 (t)

⊛ Derive an expression for Einstein's relation btwn diffusion co-efficient & ionic mobility?

⇒ From the relation btwn drift speed & the electric field strength, we know,

$$S = \frac{D F_{el}}{RT} \quad \text{------} \textcircled{1}$$

where, F_{el} = the electrostatic force per mole of ions

D = diffusion co-efficient

For an ion in solution the drift speed is,

$$S = \mu E \quad \text{----} \textcircled{11}$$

where, μ = ionic mobility

and the force on each ion in an electric field of strength is eZE. It follows that, the force per mole of ions,

$$F_{el} = N_A eZE$$

$$\Rightarrow F_{el} = ZFE \quad \text{---} \textcircled{111} \quad [\because F = N_A e]$$

where, F = faraday's constant

Now, From ①, ⑪ & ⑪①, $\mu E = \frac{DZFE}{RT}$

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$$\Rightarrow \mu = \frac{DzF}{RT}$$

This is the Einstein relation btwn diffusion coefficient & ionic mobility.

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